<$[Zr_6(\mu_3\text{-}O)_4(\mu_3\text{-}OH)_4(O_2C{-}C_6H_4{-}CO_2)_{12}]$, UiO-66

Materials
50 mL round bottom flask
Two centrifuge tubes

Chemicals
Zirconium(IV) chloride $ZrCl_4$
Acetic Acid CH_3COOH
Hydrochloric acid 37%
Benzoic Acid
Dimethylformamide (p.a.)
Terephtalic acid (H_2BDC) $C_8H_6O_4$
Ethanol (p.a.)

Synthesis
In a 50 mL round bottom flask, $ZrCl_4$ (0.235 g, 1.0 mmol) and either CH_3COOH solution (2.5 mL, 24 mmol) or HCl solution (2 mL, 24 mmol) or benzoic acid (2.93 g, 24 mmol) are dissolved in DMF (30 mL). H_2BDC is added to the clear solution (0.235 g, 1.4 mmol). The system is heated for 24 h at 120 °C.
After 24 h, the solution is cooled to room temperature and the precipitate is isolated by centrifugation. The solid is suspended in DMF (30 mL) and then recovered by centrifugation. This process is repeated twice. The obtained powder is washed with ethanol (2·30 mL) in the same way as described before (DMF washing step). Finally, the solids are dried under reduced pressure.

Analytics
- Surface area and porosity measurement using the NOVA sorption analyzer. The sample should be degassed at 120 °C for 2 h.
- IR (Infrared) spectrum. The analysis should be performed in a neat sample using a KBr disk.
- PXRD (Powder X-Ray Diffraction) phase analysis
- TGA (Thermo Gravimetric Analysis). The analysis should be performed on a fresh, non degassed sample.

Control questions
- What is the structure of UiO-66? What is the coordination environment of the zirconium atom? What is the secondary building block (SBU). What does ‘isoreticular’ means and are there ‘isoreticular’ MOFs regarding to UiO-66 ?
- The compound you obtained is ‘highly defected’ (the number of terephthalates is close to four instead of six, which is typical for the case, when no modulator is used).What kind of structural defects do you know? How can they be characterized?

Literature
1. Katz, M. J.; Brown, Z. J.; Colón, Y. J.; Siu, P. W.; Scheidt, K. a; Snurr, R. Q.; Hupp, J. T.; Farha, O. K. A Facile Synthesis of UiO-66, UiO-67 and Their Derivatives. *Chem. Commun.* 2013, *49* (82), 9449–9451.
2. F. Jeremias, V. Lozan, S. K. Henninger, C. Janiak, *Dalton Trans.* 2013, *42*, 15967.
3. J. H. Cavka, S. Jakobsen, U. Olsbye, N. Guillou, C. Lamberti, S. Bordiga, K. P. Lillerud, *J. Am. Chem. Soc.* 2008, *130*, 13850–13851.
4. M. Taddei, *Coordination Chemistry Reviews* 2017, 343, 1-24.

Vorprotokoll: UiO-66

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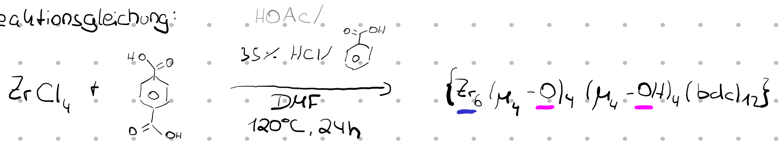
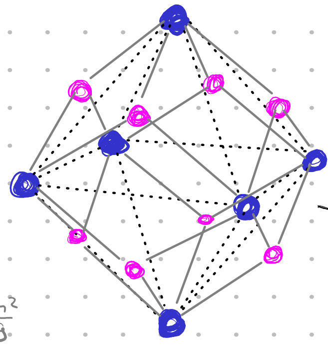
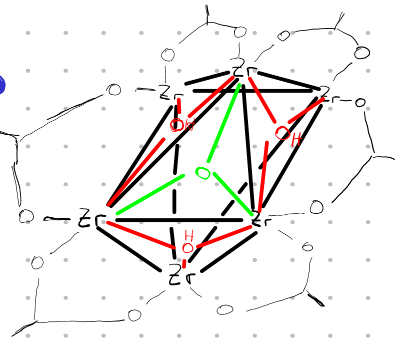
Reaktionsgleichung:







BET-Oberfläche · mit HCl: 1450 $\frac{m^2}{g}$
· mit Benzoesäure: 1242 $\frac{m^2}{g}$

Versuchsablauf:
- 0,235g $ZrCl_4$ in 50mL Rundkolben
- Zugabe 2,5mL HOAc ODER 2mL HCl ODER 2,93g Benzoesäure in 30mL DMF lösen
- Zugabe 0,235g H_2BDC
- für 24h bei 120°C erhitzen, auf RT abkühlen
- Feststoff über Zentrifugation abtrennen
- in 30mL DMF suspendieren, abzentrifugieren ×2
- " " EtOH " " ×2
- bei Unterdruck trocknen

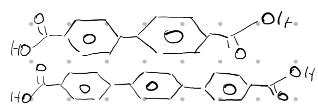
Analytik
- für Oberflächen- & Porositätsmessungen 2h bei 120°C entgasen
- IR an KBr-Plättchen messen
- PXRD
- TGA an frischer (nicht entgaster) Probe

Gefahrstoffe:

Chemikalie	Gefahrenwort	GHS-Symbole	H-Sätze	(EUH)	P-Sätze
$ZrCl_4$	Gefahr	ätzend	H302, H314	EUH014, EUH071	P280, P284, P305+351+338, P303+361+353, P402+404
HOAc	Gefahr	entzündlich, ätzend	H226, H314		P210, P280, P301+330+331, P303+361+353, P305+351+338, P310
HCl	Gefahr	ätzend, reizend	H290, H314, H335		P234, P261, P271, P280, P303+361+353, P305+351+338
Benzoesäure	Gefahr	ätzend, gesundheitsschädlich	H315, H318, H372		P260, P264, P280, P302+352, P305+351+338, P314
H_2BDC	—	—	—	—	—
DMF	Gefahr	entzündlich, gesundheitsschädlich, reizend	H226, H312+H332, H319, H360D		P210, P280, P303+361+353, P304+340+312, P305+351+338, P308+313
EtOH	Gefahr	entzündlich, reizend	H225, H319		P210, P233, P240, P241, P242, P305+351+338

$\{Zr_6\}$: 12 Nachbarn → ccp

isoreticulär ⇒ gleich gebaut: Derivate ⇒ UiO-67: bpdc Linker
UiO-68: tpdc Linker



Defekte:
- Verschiebung
- lokaler Defekt
- Defekte größeren Maßstabs

Analyse via Einkristall-XRD